

Final Project

Deep Learning in Medical Imaging

2026

1. Overview

The goal of this project is to provide you with hands on experience with deep learning algorithms in the medical field. The topic of this project is entirely up to you. Some ideas include:

- Suggesting and performing an improvement to an existing paper.
- Covering uncovered fields by a paper.
- Applying an algorithm on different tasks/applications and datasets.
- You can use algorithms that do not involve medical images and apply them on medical images to solve a specific task.
- Participating in a medical image challenge.
- Implementing by yourself a code of an existing paper (if the paper does not have a published code)

The project is 60% of your grade and is to be performed in pairs.

The project includes a project proposal, an oral presentation and a report and code submission.

2. Choosing a topic

The first step is selecting a topic of your interest. It can be done by 1) choosing a paper that interests you; 2) by choosing a task and dataset or 3) selecting a challenge in medical imaging.

I. Papers:

Please select a paper of interest from one of the recent publications in top conferences or journals listed below.

Any other requests require approval.

Conferences:

- MICCAI: Medical Image Computing and Computer-Assisted Intervention
- ISBI: IEEE International Symposium on Biomedical Imaging
- MIDL: Medical Imaging with Deep Learning
- CVPR: IEEE Conference on Computer Vision and Pattern Recognition
- ICCV: International Conference on Computer Vision
- ECCV: European Conference on Computer Vision

- NeurIPS: Neural Information Processing Systems
- ICLR: International Conference on Learning Representations

Journals:

- Medical Image Analysis
- Radiology
- Neurocomputing
- IEEE Transactions on Medical Imaging
- Nature portfolio paper (e.g. Nature npj, Nature Methods , Nature communication)

II. Tasks and datasets

Another option is to choose a medical imaging task or application. Following are some examples.

Tasks in medical imaging:

- 1) Detection- Organ/region/landmark localization.
- 2) Classification - Object or lesion classification.
- 3) Segmentation- Organ and substructure segmentation, Lesion segmentation.
- 4) Image Registration - Aligning medical images so that their information can be compared.
- 5) Content-based Image Retrieval - searching for digital images in large database.
- 6) Image Generation and Enhancement - Data generation with Generative adversarial network (GANs).

Applications in medical imaging:

- 1) Brain - Brain tumor classification/ segmentation (such as multiple sclerosis lesion segmentation), brain tracts segmentation.
- 2) Lungs - Lung tissue analysis, Nodule detection/Classification, Tuberculosis detection, Thorax-abdomen segmentation.
- 3) Liver - Tumor segmentation/ classification.
- 4) Eyes - Retinal abnormalities detection/ segmentation, eye diseases diagnosis.
- 5) Pathology and microscopy - Tissue classification, mitosis detection, skin lesion analysis, nucleus segmentation, lymph node detection and tracking.

III. Medical imaging challenge:

Another option is competing/performing a challenge in the medical Imaging field or using the dataset from one of the challenges for a different task. Some examples of challenges include:

1. Grand challenges <https://grand-challenge.org/challenges/>
2. Medical imaging challenges in Kaggle <https://www.kaggle.com>

3. Project Proposal

The project proposal should be in 1 page. Your project proposal should include:

- a. Students IDs, names, and mail addresses.
- b. What paper/papers/challenge/task/application did you choose?
- c. What is the problem that you will be investigating? Why is it interesting?
- d. What dataset will you use?
- e. What method/model will you use? Are there any existing implementations? If so, in which libraries are they written?
- f. How will you evaluate the results?
- g. What will be the innovation in your project? Though this may change while doing the project, it is important to have a goal from the beginning

Submit your proposal as a PDF file by moodle.

4. Project Presentation

Your presentation should be 12 minutes and in English. The presentation will be presented in Dates Defined below & should include the following:

- Description of your problem – what is the problem you are trying to solve? Why is it important?
- Your proposed solution and your innovation.
- Your results

5. Project Report

Your final report should be around 8 pages long, and structured like a paper (in English). Latex/lyx is recommended but not obligatory. The following is a suggested structure for your report.

- Title, Author(s), ids, e-mails.
- Abstract: Briefly describe your problem, approach, and key results. Should be no more than 300 words.
- Introduction (10%): Describe the problem you are working on, why it's important, and an overview of your results
- Related Work (10%): Discuss published work that relates to your project. How is your approach similar or different from others?

- **Data (5%):** Describe the data you are working with for your project. What type of data is it? Where did it come from? How much data are you working with? Did you have to do any preprocessing, filtering, or other special treatment to use this data in your project?
- **Methods (25%):** Discuss your approach for solving the problems that you set up in the introduction. Why is your approach the right thing to do? Did you consider alternative approaches? You should demonstrate that you have applied ideas and skills built up during the quarter to tackling your problem of choice. It may be helpful to include figures, diagrams, or tables to describe your method or compare it with other methods.
- **Experiments (20%):** Discuss the experiments that you performed to demonstrate that your approach solves the problem. The exact experiments will vary depending on the project, but you might compare with previously published methods, perform an ablation study to determine the impact of various components of your system, experiment with different hyperparameters or architectural choices, use visualization techniques to gain insight into how your model works, discuss common failure modes of your model, etc. You should include graphs, tables, or other figures to illustrate your experimental results.
- **Conclusion (5%)** Summarize your key results - what have you learned? Suggest ideas for future extensions or new applications of your ideas.
- **Writing / Formatting (5%)** Is your paper clearly written and nicely formatted?
- **Appendix (20%)** should include at least all the coding you have done. As known, most papers have a github with relevant code.

Be sure to mention all links to the code parts you have collected online.

Notice: The goal of this project is for you to gain experience in coding thus, to gain full points you have to either write an algorithm by yourself or implement major improvements/changes to an existing algorithm.

Submission

You should submit a single PDF file with the report along with your code by mail .

Instructions will be given.

6. Important Dates

- Project proposal submission – final deadline: 24.5.26 at 23:55
(can suggest earlier - and start working following approval)
- Project presentation dates: 5.7.26, 12.7.26, +possibly will schedule an additional date after end of semester
- Project report submission: 15.7.26 at 23:55

7. Examples

The required project is quite similar (though not exactly the same) to the one at: <http://cs231n.stanford.edu/project.html>. This website has many report examples, which you may look at for inspiration.

Good luck!